CHAPTER 02

LITERATURE SURVEY

2.1 Survey

The application of machine learning (ML) techniques to electroencephalogram (EEG) signals has gained significant attention in mental health research, particularly for the early detection of Major Depressive Disorder (MDD). EEG provides a non-invasive, cost-effective method for measuring brain activity, offering rich temporal data that can be analyzed to detect abnormalities associated with depression. Studies have demonstrated that ML algorithms, such as Support Vector Machines (SVM), Random Forests.

Recent advancements have focused on extracting meaningful features from EEG signals, such as power spectral density, entropy, and functional connectivity, to train ML models with high accuracy and sensitivity. Projects integrating EEG and ML show promise in supporting clinicians by providing objective tools for MDD diagnosis. Despite progress, challenges persist, including variability in EEG signals across individuals, lack of large and balanced datasets, and limited generalization across demographics. This project builds upon these studies by aiming to develop a scalable and accurate ML-based system that utilizes EEG features for early MDD detection, addressing issues of data variability and focusing on robust preprocessing and feature engineering strategies.

2.2 Motivation

The motivation for this project arises from the growing need for accurate and early detection of Major Depressive Disorder (MDD) using advanced artificial intelligence techniques. MDD is a leading cause of disability worldwide, and its diagnosis is often based on subjective clinical assessments, which can result in delays and inconsistencies in treatment. With advancements in computational psychiatry, integrating machine learning with electroencephalography (EEG) data offers a promising approach to improving diagnostic accuracy. EEG provides a non-invasive method to analyse brain activity, and machine learning models can effectively detect patterns associated with MDD. By leveraging techniques such as feature extraction, data augmentation, and classification algorithms, this project aims to build a robust system that can classify individuals as either MDD patients or health controls. The objective is to enhance early detection, facilitate timely intervention, and contribute to more reliable and scalable mental healthcare solutions.

2.3 EEG-Based Classification for Depression

Several studies have utilized machine learning models to classify EEG data for depression detection. Hosseini Fard et al. (2013) applied feature extraction techniques like wavelet transform and used classifiers such as SVM and k-NN, achieving high classification performance. Similarly, Mumtaz et al. (2017) applied a hybrid feature selection approach to improve the SVM model's ability to detect depression. These studies confirm the viability of EEG-based classification frameworks using ML for early MDD diagnosis.

2.4 Deep Learning Approaches

Recent developments have seen the application of deep learning, particularly CNNs and Long Short-Term Memory (LSTM) networks, to EEG-based depression detection. Zeng et al. (2019) employed a deep CNN model trained on EEG spectrograms, reporting superior performance compared to traditional ML models. Likewise, Zhang et al. (2020) used a hybrid CNN-LSTM model to capture both spatial and temporal EEG features, significantly enhancing depression classification accuracy. These studies underscore the potential of deep learning to automate feature extraction and improve classification outcomes.

2.5 Feature Engineering and Channel Selection

Optimizing EEG channel selection and feature engineering has been critical in improving model performance. Acharya et al. (2015) emphasized the importance of selecting relevant EEG channels and time-frequency features, showing that reduced channel sets could still yield high classification accuracy. Furthermore, Lee et al. (2011) demonstrated that alpha and theta band power in the frontal and parietal regions were strongly correlated with depressive symptoms, guiding feature selection in ML-based models. These findings show that intelligent preprocessing and feature optimization are key to effective EEG-based

2.6 Challenges and Opportunities

Despite promising results, several limitations exist in current EEG-ML systems for MDD detection. Data scarcity, class imbalance, and inter-subject variability often affect model generalizability. Mohammadi et al. (2019) highlighted the need for larger, multi-centre datasets and standardized EEG acquisition protocols. Additionally, interpretability of ML models remains a challenge, particularly in clinical settings. Future research must focus on explainable AI (XAI) approaches and rigorous validation to ensure trust and clinical adoption. This project aims to address these gaps by building a generalized and interpretable ML framework for early MDD detection using standardized EEG protocols.